Calibrated and Kernel-Based Detection of QKD Implementation Attacks: A Controlled Simulation Study

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***Abstract*—Machine-learning detectors for quantum key distribution (QKD) implementation attacks are usually evaluated on simulated telemetry, and frequently report near-perfect accuracy. We show that such numbers can be an artifact of the simulator: without operating-condition drift, a single telemetry feature separates attacks from normal blocks perfectly. We build a seeded decoy-state BB84 telemetry simulator with drift in loss, misalignment, detector efficiency and background light, finite-size QBER noise, and a single *attack-strength* parameter s that scales four attack signatures (photon-number splitting, detector blinding, Trojan horse, intercept-resend) from blatant to subtle. On it we compare a Laplace Gaussian-process classifier (GP), an amplitude-encoding fidelity-kernel SVM (AE-SVM, 3 qubits, computed exactly and verified against a PennyLane circuit to 10^{-14}), RBF-SVM, random forests and a GP+AE-SVM stacking model, all at matched training sizes over 10 seeds. At $s = 0.3$, every supervised model lands between 0.80 and 0.84 MCC; random forest is best or tied-best throughout, and the GP is neither the most accurate nor the best calibrated (ECE 0.047 vs. 0.030 for isotonic-calibrated RF), so it is not the best basis for uncertainty-based deferral. The amplitude kernel does carry a usable inductive bias: it wins on direction-only control tasks and, on unseen attack types, matches RBF-SVM—but only because one feature (a Mahalanobis distance) carries the input's magnitude. Removing that feature drops AE-SVM's zero-day AUC from 0.95 to 0.55, while appending a constant "norm anchor" coordinate before encoding restores it to 0.94. Code, data and every result file are released.**

***Index Terms*—Quantum key distribution, anomaly detection, Gaussian processes, quantum kernels, calibration, simulation methodology**

I. Introduction

QKD derives its security from physics [1], [2], but deployed systems leak through implementation side channels: photon-number splitting (PNS) against multi-photon pulses [3], [4], detector control by bright illumination [5], [6], Trojan-horse probing of the transmitter or receiver [7], [8], and intercept-resend [9]. Decoy states [10] and measurement-device-independent designs [11] close some of these channels; continuous telemetry monitoring is a complementary defense, and machine learning has been applied to it [12] and to related QKD control problems [13].

Two methodological problems make it hard to know what such detectors are worth. First, they are typically trained and tested on simulated telemetry, and a simulator in which normal operation sits at one fixed operating point makes almost any attack signature separable. Second, "quantum" components are often evaluated at very different training sizes from classical baselines, so a comparison confounds model and data.

This paper is a controlled simulation study that addresses both. Our contributions are:

1) A seeded decoy-state BB84 telemetry simulator with operating drift and an explicit attack-strength axis s, together with leakage checks. Without drift, a single feature gives perfect separation; with it, detection degrades smoothly from MCC ≈ 0.97 at $s = 1$ to ≈ 0.5 at $s = 0.05$ (Section III).
2) A matched-n comparison of a GP, an exact amplitude-encoding fidelity kernel, RBF-SVM, random forest (RF) and stacking at every training size from 25 to 1400 (Section V).
3) A mechanism for when the amplitude kernel generalizes to unseen attacks, confirmed by ablation: it needs a feature that carries the input's magnitude, which can be supplied by a one-line norm anchor (Section V-D).
4) A calibration and selective-prediction comparison showing that the GP's posterior is not the best uncertainty signal on this data (Section V-F).

All results are from simulation; Section VI states what this does and does not establish.

II. Related Work

Hardware countermeasures monitor detector statistics directly, for example after-pulse and efficiency estimates computed in real time [14]. Mao et al. [12] trained neural networks to recognise attacks on continuous-variable QKD from measured statistics. We study discrete-variable decoy-state BB84 telemetry and focus on evaluation methodology, calibrated uncertainty and kernel inductive bias rather than on a new detector.

Quantum kernel methods embed data in a Hilbert space through a state-preparation circuit and use state overlaps as a kernel [15], [16]. Provable advantages exist for specially constructed problems [17], and whether a quantum kernel helps

on a given dataset depends on how its geometry matches the data [18]. Amplitude encoding is a case in which the overlap is classically computable (Proposition 1), so it isolates the kernel's inductive bias from any question of quantum speed-up. Gaussian processes [19] and other Bayesian methods [20] provide model-based uncertainty for anomaly detection; post-hoc recalibration [21]–[23] is the standard alternative for point-estimate models.

III. Telemetry Simulator

A. Channel model

Each sample is one key-generation block of a decoy-state BB84 link over $L = 50\,\text{km}$ of fiber ($\alpha = 0.2\,\text{dB/km}$). For mean photon number $\mu \sim \mathcal{U}(0.1, 1)$, transmittance $T = 10^{-A/10}$ and detector efficiency η, the gain and QBER of a Poisson source are [10]

$$Q_\mu = Y_0 + \left(1 - e^{-\eta T \mu}\right)(1 + p_{\text{ap}}), \tag{1}$$

$$E_\mu = \left[\tfrac{1}{2} Y_0 + e_d \left(1 - e^{-\eta T \mu}\right)(1 + p_{\text{ap}})\right] / Q_\mu, \tag{2}$$

with afterpulse probability $p_{\text{ap}} = 0.005$. Normal operation is not a single point: per block, the attenuation A drifts by $\mathcal{N}(0, 0.3^2)\,\text{dB}$ around αL, η by 3% around 0.9, and the misalignment error e_d, dark-count probability Y_0 and background rate are log-normal around 0.01, 10^{-5} and $100\,\text{s}^{-1}$ (log-scale s.d. 0.4, 0.3, 0.3). The two reported error rates (sifted QBER and decoy-basis error rate) carry independent finite-size noise $\sqrt{E(1-E)/N_b}$ with $N_b = 5000$. Seven measurements are reported per block: QBER, photon count rate, detection efficiency, dark-count rate, error rate, attenuation and background rate.

B. Attack models and strength

The four attacks perturb these measurements as in prior qualitative descriptions [3], [5], [7], [9]: PNS removes a fraction 0.3 of photons and raises QBER by $\mathcal{U}(0.02, 0.08)$; blinding divides efficiency by $1 + 1.5\gamma$, $\gamma \sim \mathcal{U}(0.5, 1.5)$, and raises QBER by $\mathcal{U}(0.05, 0.25)$; Trojan-horse probing multiplies background by $1 + 10\rho$, $\rho \sim 0.05\,\mathcal{U}(0.5, 2)$; intercept-resend raises QBER by $\mathcal{U}(0.10, 0.35)$ and lowers the count rate. These magnitudes are modeling assumptions, not measured values. Every deviation from the clean block is multiplied by a strength $s \geq 0$, so $s = 1$ is the default and $s \to 0$ makes an attack indistinguishable from normal operation (Fig. 1).

C. Leakage checks

Two properties are tested in the released test suite. (i) Different seeds give different datasets, and every random draw, including attack perturbations, derives from the dataset seed. (ii) No single feature is a label proxy: block timestamps are drawn independently of the label, and a depth-2 decision tree on any single feature stays below perfect accuracy. We include these checks because an earlier version of our own pipeline failed both, which made every detector look near-perfect; we expect the same failure in other simulation-only QKD detection results.

TABLE I
Detection at $n = 1400$ (mean ± 95% CI, 10 seeds). ECE uses 15 equal-width bins.

	$s = 1.0$			$s = 0.3$		
Model	MCC	AUC	ECE	MCC	AUC	ECE
RF	**.975**±.008	.999	.013	**.834**±.013	.969	.030
RF-isotonic	.972±.008	.997	**.010**	**.837**±.013	.967	**.030**
Hybrid	.971±.005	.999	.015	**.833**±.016	.969	.037
RBF-SVM	.969±.008	.999	.011	.810±.015	.957	.040
GP	.964±.009	.999	.039	.801±.016	.956	.047
AE-SVM	.955±.011	.999	.015	.814±.023	**.970**	.032
OC-SVM	.838±.016	.994	–	.657±.017	.839	–
IF	.837±.021	.997	–	.722±.019	.929	–
LOF	.824±.021	.997	–	.691±.011	.924	–

IV. Detectors and Protocol

Features. Every detector sees the same 8 features (Fig. 2): the 7 measurements standardized on the training split, plus the robust Mahalanobis distance [24] of the standardized vector to the training-split normal blocks.

Amplitude-encoding fidelity kernel (AE-SVM). A vector $\mathbf{x} \in \mathbb{R}^d$ is zero-padded to 2^q entries, $q = \lceil \log_2 d \rceil$, and normalized, giving the state $|\psi(\mathbf{x})\rangle = \sum_i \hat{x}_i |i\rangle$. The fidelity kernel is $K_A(\mathbf{x}, \mathbf{y}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{y}) \rangle|^2$.

Proposition 1: $K_A(\mathbf{x}, \mathbf{y}) = (\hat{\mathbf{x}}^{\mathsf{T}} \hat{\mathbf{y}})^2$, the squared cosine similarity of the inputs. It is computable classically in $\mathcal{O}(d)$ time per pair and is invariant to $\mathbf{x} \mapsto c\,\mathbf{x}$ for every $c \neq 0$, including $c = -1$.

The proof is immediate from real amplitudes and zero padding. We therefore evaluate the kernel exactly with all 8 features on 3 qubits, and confirm that a PennyLane [25] state-vector simulation of the overlap circuit agrees to 7×10^{-15}; the circuit costs 26 ms per kernel entry versus 3×10^{-4} ms in closed form. The SVM uses $C = 1$, balanced class weights and Platt-scaled probabilities [21].

Baselines. A Laplace GP classifier [19] with an RBF+white-noise kernel (hyperparameters chosen by marginal likelihood on a stratified 300-sample subset, then fixed for the full fit); RBF-SVM with the same SVM settings as AE-SVM; RF (200 trees) [26], raw and isotonic-calibrated [22]; a stacking model (Hybrid) combining GP and AE-SVM through logistic regression with 3-fold cross-validation [27]; and three novelty detectors fit on normal training blocks only (one-class SVM, isolation forest, LOF).

Protocol. $N = 2000$ blocks with 30% attacks (balanced across types), 70/30 stratified split, 10 seeds; we report means with 95% confidence intervals over seeds. Matched-n experiments train *every* model on the same stratified subsample, for $n \in \{25, 50, 100, 200, 500, 1000, 1400\}$. Paired model comparisons use the exact McNemar test per seed. Zero-day experiments train on three attack types and test on normal blocks plus the held-out fourth, at 1,200 test blocks per split.

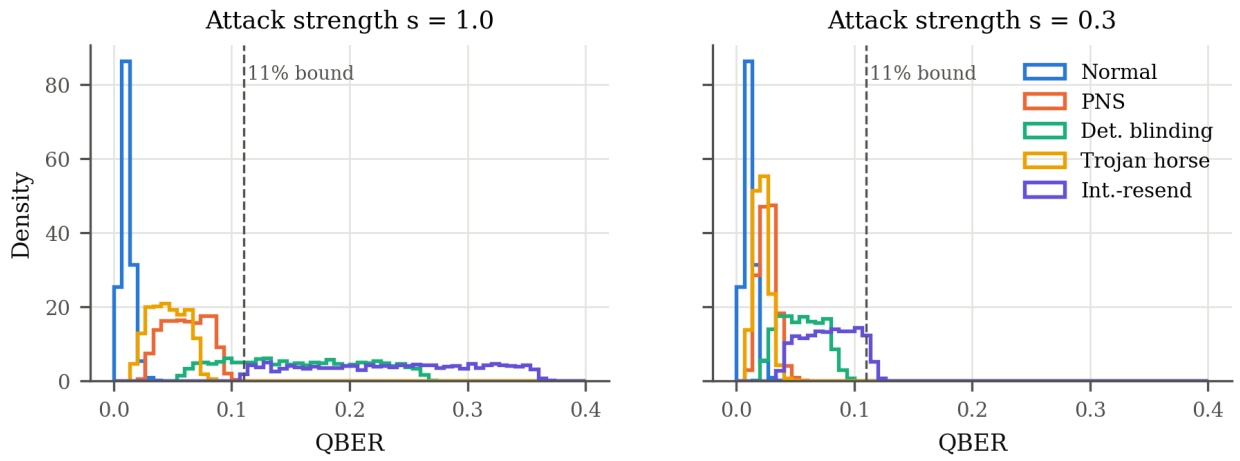


Fig. 1. QBER by class at attack strength $s = 1$ and $s = 0.3$. At $s = 0.3$, PNS and Trojan-horse blocks sit well below the 11% bound and overlap normal operation.

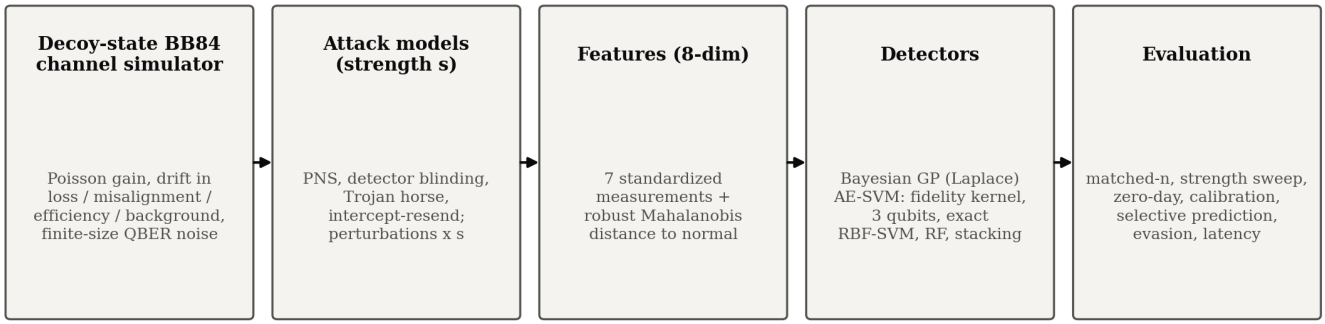


Fig. 2. Study design: simulator, attack models with strength s, a fixed 8-dimensional feature map shared by every detector, and the evaluation suite.

V. Results

A. In-distribution detection

Table I reports detection at the full training size. At $s = 1$ every supervised model exceeds 0.95 MCC and 0.99 AUC, so this operating point does not discriminate between methods; the novelty detectors, which never see an attack label, reach 0.82–0.84 MCC. At $s = 0.3$ the supervised models spread between 0.80 and 0.84 MCC: RF, isotonic RF and Hybrid lead, and the GP has the lowest attack recall (0.78 ± 0.02, against 0.85 for RF and 0.90 for AE-SVM) because it trades recall for specificity (0.98). Hybrid improves on the GP alone (pooled over seeds, Hybrid corrects 154 GP errors and introduces 76; McNemar $p < 0.05$ on 4 of 10 seeds) but does not improve on RF. AE-SVM matches RBF-SVM in accuracy (both 0.920; McNemar $p > 0.05$ on every seed) and has the highest AUC (0.970).

B. Matched training size

Fig. 3 trains every model at every n. RF is the most sample-efficient model at both strengths (MCC 0.75 at $n = 25$, $s = 0.3$, against 0.65–0.68 for the kernel methods and GP). The amplitude kernel needs more data than the RBF kernel to reach the same MCC: at $s = 0.3$ AE-SVM trails RBF-SVM by 0.03–0.04 for $n = 50$–200 and reaches parity only from $n = 1000$ (0.804 vs. 0.806); at $s = 1$ it trails by 0.008–0.017 at every $n \geq 50$. There is no training size at which it is reliably ahead.

C. Attack strength

All four detectors degrade together as s falls (Fig. 4), from MCC 0.96–0.98 at $s = 1$ to 0.61–0.62 at $s = 0.1$ and 0.47–0.51 at $s = 0.05$. RF is best at every $s \geq 0.2$; below that all four are within 0.04 MCC of each other. Per attack type, intercept-resend and detector blinding keep GP recall ≥ 0.97 down to $s = 0.1$ because they move QBER the most, while PNS and Trojan-horse probing are lost first (Trojan-horse recall 0.45 at $s = 0.3$).

D. Unseen attack types

In leave-one-attack-out tests (Fig. 5) AE-SVM generalizes as well as RF and better than RBF-SVM at $s = 0.3$ (mean AUC 0.952 vs. 0.917), despite a lower kernel-target alignment [28] with the labels (0.13–0.22 vs. 0.29–0.66 across

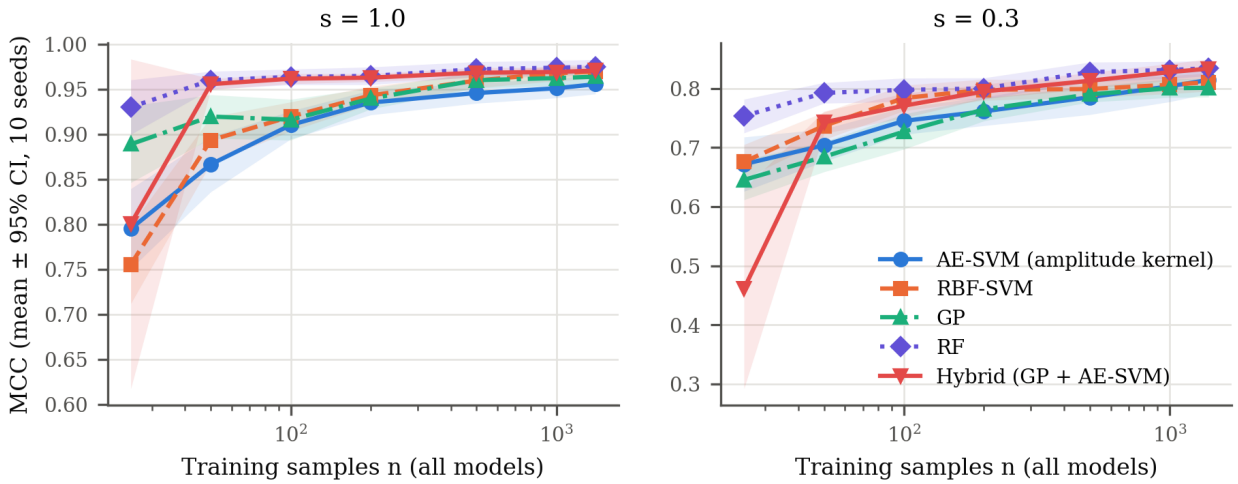


Fig. 3. MCC versus training size, every model trained on the same stratified subsample (mean ± 95% CI, 10 seeds). The Hybrid model is unstable at $n = 25$, where its 3-fold stacking has only ≈8 training blocks per fold.

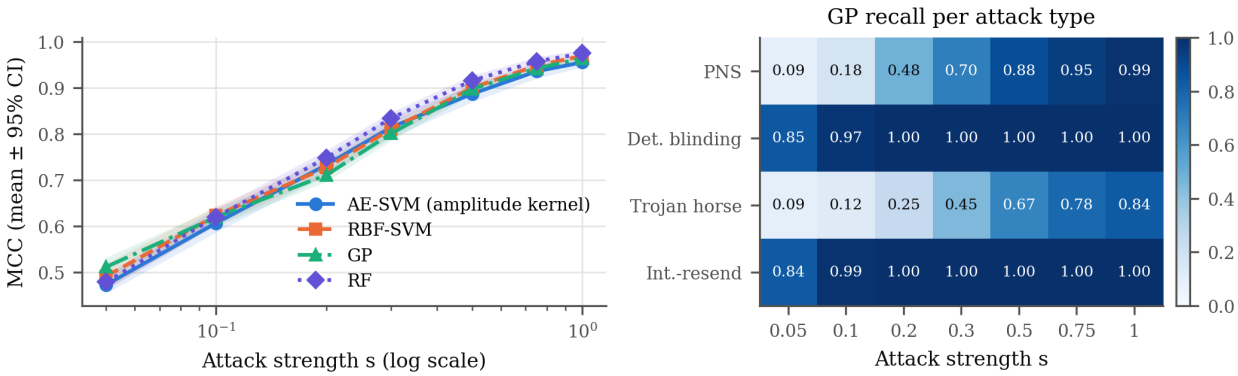


Fig. 4. Left: MCC versus attack strength at $n = 1400$. Right: GP recall per attack type. Trojan-horse and PNS signatures are lost first.

attack types). Proposition 1 suggests why this is fragile: K_A ignores $\|\mathbf{x}\|$, and an unseen attack is most easily recognised by *how far* it is from normal operation, which is a magnitude. In our feature map that magnitude is carried by one coordinate, the Mahalanobis distance. Removing it (Fig. 6) drops AE-SVM's mean zero-day AUC to 0.554 at $s = 0.3$ and 0.635 at $s = 1$, while RBF-SVM is unaffected (0.940, 0.987). Appending a constant coordinate $c = \text{median}_{\text{train}}\|\mathbf{x}\|$ before encoding, so that $\phi(\mathbf{x}) = (\mathbf{x}, c)/\|(\mathbf{x}, c)\|$ retains the input's norm relative to c, restores AE-SVM to 0.939 and 0.980 without the Mahalanobis feature.

E. When the amplitude kernel helps

Fig. 7 isolates the inductive bias. When the label depends only on direction and the norm is a heavy-tailed nuisance, AE-SVM learns from far fewer samples than any baseline (0.71 vs. 0.56 for RBF-SVM at $n = 25$; 0.97 vs. 0.74 at $n = 1000$). When the label depends only on norm, AE-SVM is at chance at every n. QKD telemetry is neither extreme, which is why the kernel is competitive but not superior on it.

F. Calibration and selective prediction

The GP's Laplace posterior is the least calibrated supervised model here (ECE 0.047 ± 0.006 at $s = 0.3$ and 0.039 ± 0.002 at $s = 1$), and isotonic RF the best (0.030 and 0.010). For selective prediction (Fig. 8), the GP's probabilities are concentrated near the decision boundary: at threshold $u < 0.3$ it keeps 69% of test blocks at 97.6% accuracy, while isotonic RF keeps 80% at the same accuracy and AE-SVM 79% at 97.8%; at $u < 0.1$ the GP retains no blocks at all at $s = 0.3$. A model-based posterior is attractive in principle, but on this data a cheaply recalibrated point estimator gives the better deferral policy.

G. Evasion and latency

A simple gradient-free evasion that moves each attack block's five most important features toward the normal median (Fig. 9) lowers RF and GP detection together, with RF ahead by 2–9 points at every budget at $s = 0.3$ (2–10 points at $s = 1$). The GP's mean uncertainty on evading blocks barely changes (0.30 to 0.33 at $s = 0.3$), so uncertainty alone does

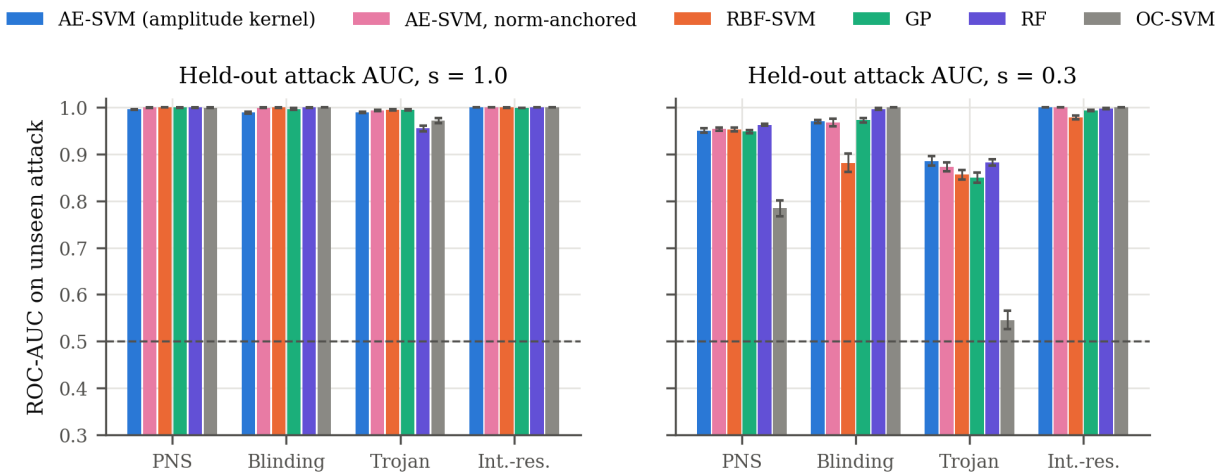


Fig. 5. AUC on a held-out attack type, training on the other three (mean ± 95% CI, 10 seeds).

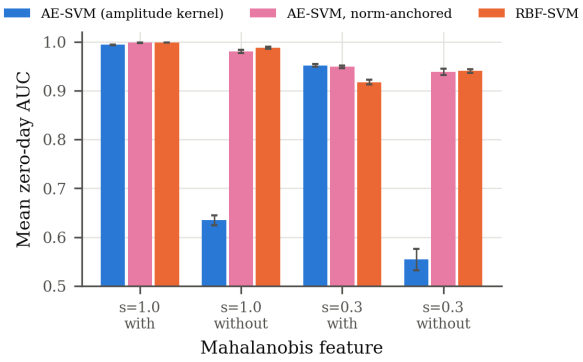


Fig. 6. Mean zero-day AUC with and without the Mahalanobis feature. Without it, the amplitude kernel discards all magnitude information and fails on unseen attacks; a norm anchor restores it.

TABLE II
INFERENCE LATENCY ON ONE CPU CORE (TRAINING EXCLUDED). BATCH: AMORTISED OVER 600 BLOCKS. SINGLE: MEDIAN FOR ONE BLOCK, DOMINATED BY PYTHON CALL OVERHEAD.

Component	Batch (ms/block)	Single (ms)
Feature map	0.0004	0.14
AE-SVM (exact)	0.008	0.35
RBF-SVM	0.012	0.16
RF (200 trees)	0.093	23.4
Hybrid	0.33	26.9
GP	0.38	29.6

not flag this evasion. Table II gives inference cost: all detectors are far below a per-block budget measured in seconds; the kernel SVMs are cheapest in batch, and the GP and Hybrid models are slowest.

VI. LIMITATIONS

Every result is from simulation. The attack magnitudes are assumptions, and the drift model, though grounded in standard channel physics, is not fitted to a deployed link; the strength axis s shows how conclusions move with signature size but does not say which s a real attacker produces. The evasion attack is gradient-free and not adaptive. The detectors see per-block summaries only; time-series models over consecutive blocks may do better. Validating the ranking of methods, and the zero-day mechanism, on telemetry from a physical QKD testbed is the necessary next step.

VII. CONCLUSION

On a QKD telemetry simulator designed not to be trivially separable, random forests are the strongest and most sample-efficient detector, isotonic calibration gives better uncertainty for deferral than a GP posterior, and the amplitude-encoding fidelity kernel—computable exactly on classical hardware—is competitive but not superior. Its one distinctive behavior is explained by Proposition 1: it generalizes to unseen attacks only when some feature carries the input's magnitude, and a norm anchor supplies that when no such feature exists. Simulation-based QKD detection studies should report single-feature separability and an attack-strength sweep alongside headline accuracy.

CODE AND DATA AVAILABILITY

Simulator, models, experiment driver (`scripts/hyqad/run_experiments.py`), figure script and every result CSV are at https://github.com/arnavk23/quasar-security.

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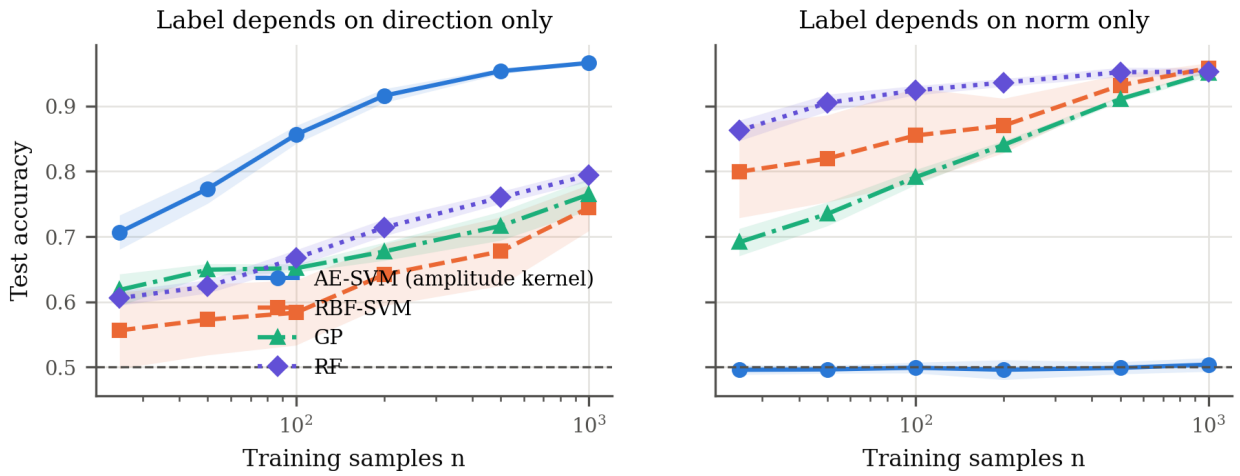


Fig. 7. Control tasks in $\mathbb{R}^8$ with heavy-tailed norms. Left: label depends on direction only. Right: label depends on norm only.

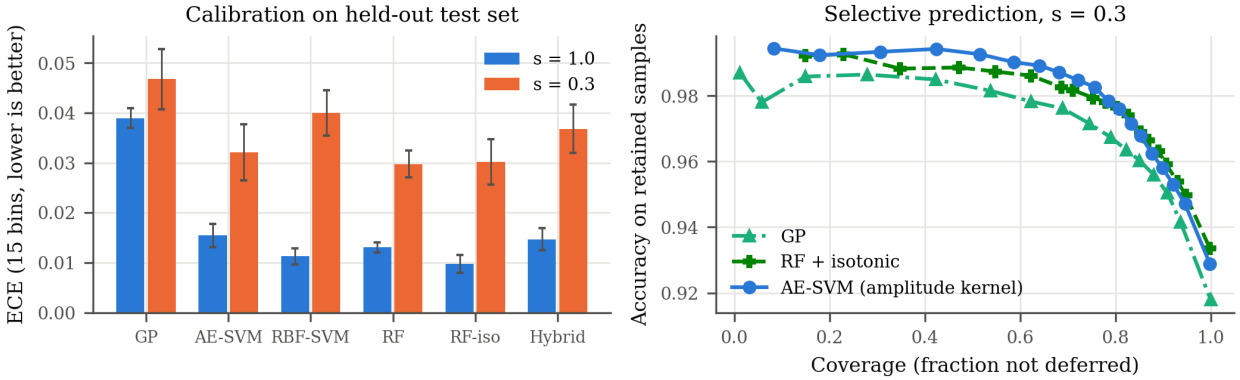


Fig. 8. Left: ECE by model. Right: accuracy versus coverage when deferring samples with uncertainty $u = \sqrt{p(1-p)}$ above a threshold ($s = 0.3$).

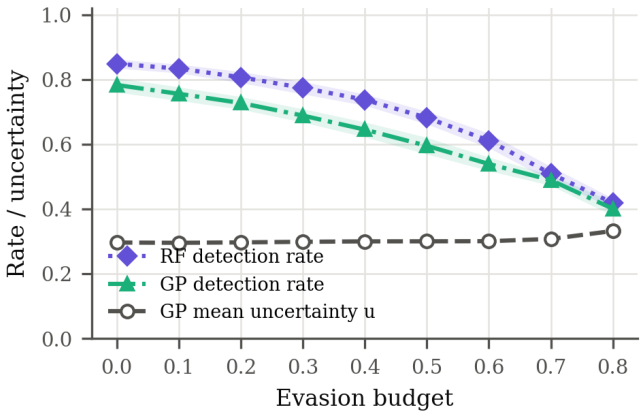


Fig. 9. Detection rate of attack blocks ($s = 0.3$) whose five most RF-important features are moved a fraction (the budget) of the way toward the normal median, and the GP's mean uncertainty on them.

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